

Instructor: Date:

Learner: Regional Branch:

Program: Center:

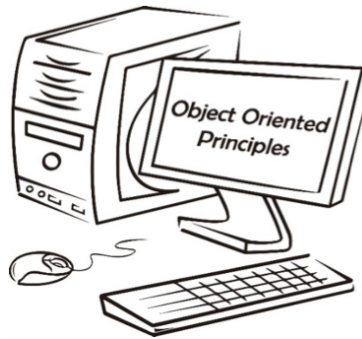
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Let's Start!

Principles in OOP



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Build the macro system conceptual model following the customer requirements, through the use and interpretation of collected information, represented in class diagrams, interaction diagrams, and operation contracts.

Can Do Statements:

- I can list the principles in OOP.
- I can talk about the characteristics for principles in OOP.
- I can discuss simple ideas in real life that relate to the OOP principles.

Ask your partner:

- Why do you think there are different types of software?
- Would you like to know essential collections of software for managing programming packages? Why?

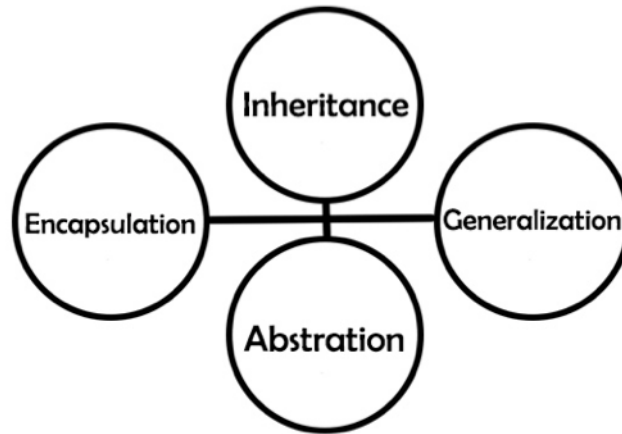




Let's Explore!

This professor wants to talk about OOP* principles.

*"OOP" means Object Oriented Programming



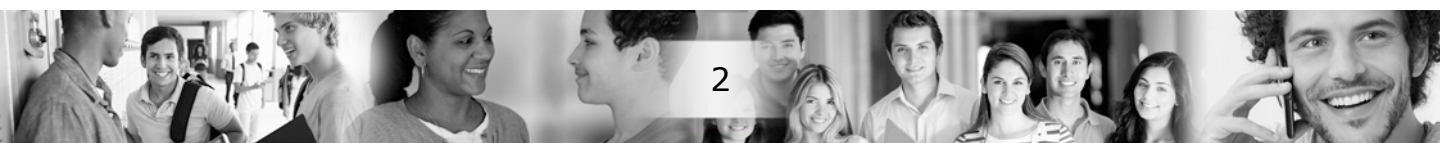
Instructions: Read the text below.

Tonight, I want to inform you about those main OOP principles. I want you to analyze this graphic; it can help us to understand how these principles work. Abstraction: This principle makes us consider relevant information about something without becoming bogged down in detail; in other words, this principle helps us ignore irrelevant details. For example, if you want to get a new computer with a different operating system; surely, you want to know those main aspects about interfaces or software security, but you don't need to know how the screen works.

Now, let's pretend you want to change not only your operating system but your computer. To do this, you may want to know how to operate your new computer, thus without being versed about every single component of the computer. This fact represents the encapsulation principle because it allows you to know how something works without worrying about its technical details.

Generalization and specialization are OOP principles that work together; I want you to pretend you need to buy a new computer and you are going to choose between certain kinds of machines such as desktops or laptop computers. This fact represents the generalization principle because computers are certainly generalized under the same category. Now, even when these objects are in the same category, they have specificities and different versions. This is specialization, a principle that establishes that machines inherit characteristics of the generalized category.

The last principle is polymorphism. It doesn't matter what computer you want to buy because you know how to interact with it already. Machines share common handling procedures, so you don't need special training to operate any types of computers.





Let's Practice!

The Principles of Generalization and Specialization in a Real Life

Instructions: Select the correct words to complete the sentences.

1. Generalization and specialization, _____ principles can work together. (Hint: Plural near you)
a. these
b. this
c. that
2. _____ objects exemplify the principle of generalization because they are part of a same category. (Hint: Plural far from you)
a. This
b. That
c. Those
3. But even if _____ is in the same category, it has specificities and different versions. (Hint: Singular close to you)
a. this
b. these
c. those
4. So, _____ refers to specialization. (Hint: Singular far from you)
a. these
b. that
c. those
5. _____ last principle is polymorphism. (Hint: Singular close to you)
a. This
b. Those
c. These
6. The principle of polymorphism is simple. _____ computers that you choose are easy to interact with. (Hint: Plural far from you)
a. This
b. Those
c. That



Let's Practice!

Time to Find and Memorize the OOP Principles

Instructions: Match column A with column B.

COLUMN A

1. This OOP principle explains how to operate something without having to know its technical parts.
2. This principle implies that machines share certain characteristics with a predecessor.
3. The ability to interact with an object irrespective of its kind is the main point of this postulate.
4. This is the principle that works together with generalization.
5. This principle claims that those technical specifications of the systems are not necessary for the user to know in order to operate a machine

COLUMN A

- a. Encapsulation
- b. Inherit
- c. Polymorphism
- d. Specialization
- e. Encapsulation



Extension Activity

OOP Principles in a Conceptual Map

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find out more about the OOP principles that you learned.
- Create a conceptual map in which you explain the principles.
- Share your conceptual map with your partner.

Self-Assessment

Yes

No

- I research about OOP principles.
- I design a conceptual map to explain the principles.
- I share my conceptual map with others.

☐	☐
☐	☐
☐	☐



Target Words

- **polymorphism** (noun): ability to interact with an object irrespective of its specialized category
e.g. The last principle is polymorphism. It doesn't matter what computer you want to buy because you know how to interact with it already.
- **desktop** (noun): computer you use on a desk
e.g. I want you to pretend you need to buy a new computer and you are going to choose between certain kinds of machines such as desktops or laptop computers.
- **inherit** (verb): to have characteristics from the predecessor
e.g. This is specialization, a principle that establishes that machines inherit characteristics of the generalized category.
- **bogged down** (verb): to be overload
e.g. This principle makes us consider relevant information about something without becoming bogged down in detail.
- **encapsulation** (noun): OOP principle that explains you how to operate something without knowing its technical parts
e.g. This fact represents the encapsulation principle because it allows you to know how something works without worrying about its technical details.



Learning Points

Review of Demonstrative Pronouns in Affirmative Statements

You use **demonstrative pronouns** to point at or identify specific things within a sentence.

Demonstrative pronouns can be singular or plural.

You use **demonstrative pronouns** depending on how near or far are the things you want to point or identify.

	Singular	Plural
Near	<i>This</i>	<i>These</i>
Far	<i>That</i>	<i>Those</i>

Examples

- *I want you to analyze this graphic.*
- *Surely, you want to know those main aspects about interfaces or software security, but you don't need to know how the screen works.*



Learning Points

Verbs With Infinitives in Affirmative Statements With 'Want To'

You use **verbs** and **infinitives** in **affirmative** statements with **want to** to express that you desire or wish something.

You need to be careful when writing a **verb** and an **infinitive** because you don't use **to** as a common preposition.

You use the following structure when writing a **verb** and **infinitive** in affirmative statements with **want to**.

Subject + **Want to** + Verb in infinitive

Example:

*I **want to add** specific data to the system.*

Examples

- *I want to inform you about those main OOP principles.*
- *Let's pretend you want to change not only your operating system but your computer.*

Understanding Abstraction in OOP

You use **abstraction** when you want to consider some complex ideas, but you don't want to focus on irrelevant details.

Let's imagine the following situation:

You want to swap your personal computer for a week with a friend.

You probably want to give him some essential information about your computer such as what operating system it uses, what kind of battery charger it has, or maybe what your administration password is.

However, you don't want to give him irrelevant details about your computer such as the internet explorer you use or where to find the calculator.

Abstraction gives you only the important facts without telling you the irrelevant details about something.

